

Tertill

Goal: \$120,000

Deadline: July 13, 2017

The Roomba's outdoorsy cousin is coming to town soon and it will take care of all your weeding needs. From the maker of Roomba, Joe Jones of Franklin Robotics (No, not the Jonas brother), this robot is waterproof, weatherproof and completely solar-powered. It's effective even on cloudy days! It's completely chemical free and child friendly too. It connects to your phone through Bluetooth to keep track of your garden conditions and also the Tertill's activities using the dedicated app.

Tertill works on a simple mechanism. Using sensors similar to those on a phone, it assesses its surroundings. Assuming that weeds are short (less than an inch) and plants are not, after the sensor touches something tall, it turns away. A shorter plant on the other hand will go under the Tertill, activate a different sensor and gets chopped down by the whacker which is a nylon string revolving at high speeds. The clippings fall back on the ground and the energy goes back into the soil over a period of time. Plant collars are provided so that the Tertill doesn't kill the saplings. If the



weed keeps coming back, Tertill will keep chopping it down until it eventually runs out of energy (talk about persistence!).

The 4-wheel drive has its wheels negatively cambered (base outwards) to ensure that pre-emerging weeds are destroyed when the garden is scrubbed. This also helps it to manoeuvre various terrains and slopes. It's smart enough to switch off a motor to prevent damage if it's stuck. The robot is designed to typically cover an area of 100 sq. Feet. For a bigger garden, multiple Tertills can be programmed to co-ordinate together and get the job done though some sort of boundary at least 2 inches tall is needed to keep the Tertill from wandering.

It has a micro-USB charging port for back-up. It removes the monotony and frustration of weeding for gardeners and farmers alike and you could maybe even have it as your new pet!

MICRO

Goal: £15,500

Deadline: June 30 2017

The jack of all adapters is getting kick started and it goes by the name of MICRO. Touted to be the world's smallest universal travel adapter with surge protection, it is compatible in over 150 countries and all 7 continents. An ergonomic design that is also very stylish, it is meant to work with all of your low power consuming devices (laptops, phones, tablets, etc.). It's very sleek, simple and ultra light (weighs a featherly 40g). Its modularity also adds to its versatility in addition to fitting in your pocket. The casing effectively covers any 3rd pin, so there is no danger of exposed plug pins. A spike which lasts for about a 1000th of a second can cause devastating damage to your valuable

devices but fear not; the MICRO's replaceable 6.3A integrated fuse will take the damage. Made of fire-retardant ABS, brass and eco-friendly silicone - the makers claim that at the time of release it will be FCC, CE and IEC certified.



The only drawbacks are that it is not grounded which means that it cannot support heavy appliances like hair dryers rated above 1700W which is outside the MICRO's range (250-1575W) and that it does not convert voltage. The sturdiness of the device also is yet to be determined.

Magpie: Smart global GPS Tracker

Goal: \$10,000

Deadline: June 29, 2017

Magpie is a smart GPS device that keeps track of your loved ones and your valuables. With tracking in 185 countries, it has a very wide coverage. The device is dirt proof and waterproof (they're aiming for an IP67 rating). It comes with a hook and a clip to use in different situations. The battery takes only an hour to charge completely. By using GPS instead of Bluetooth, they eliminate the range limit - distance is not an issue. It's easy to hide too, so you can stash it into your kid's bag and not worry about it being



discovered. Charging takes place through a micro-USB port and the case is made out of polypropylene which is ultrasonically welded.

On a single charge, the Magpie will last anywhere between 1 day to 3 months depending on the tracking frequency assigned to a particular device (once a minute, hourly, daily, etc.). It also has a geo fence alert i.e. if a particular unit goes out of a pre-designated zone (Safe Zone), you will be alerted instantly. It has a nano accelerometer built in to warn you if anything that shouldn't be moving is in motion. The makers are offering the devices for free while charging a modest \$5/month as a subscription fee for the tracking service. The creators claim to deliver new products to existing customers completely free of charge and that subscription charges will never increase, if anything they will only decrease due to the increase in customers. Which is great news if you're looking for an easy way to track stuff.